

QoE-Aware DRL for Slice Admission Control: A Live Open RAN Testbed Validation

Manoj Prasad Kunasegran

School of Engineering

Taylor's University

Subang Jaya, Malaysia

orcid.org/0009-0005-3169-1873

Wai Leong Pang*

School of Engineering

Taylor's University

Subang Jaya, Malaysia

orcid.org/0000-0001-8407-5648

Corresponding author

Swee King Phang

School of Engineering

Taylor's University

Subang Jaya, Malaysia

orcid.org/0000-0002-7877-8766

Abstract—Building on our previous work on Deep Reinforcement Learning (DRL)-driven Slice Admission Control (SAC), which was validated entirely in simulation, this paper moves the same control problem onto a real Open RAN testbed. We deploy a Quality of Experience (QoE)-aware Deep Q-Network (DQN) admission policy on a real OpenAirInterface (OAI) gNodeB (gNB) across three network slices (eMBB, URLLC, mMTC), and compare it against a static baseline and a best-possible static-at-cap policy. Results demonstrate that DQN sustains 100% SLA compliance across 45 live evaluation episodes, compared to 73.6% for the static baseline, and that this advantage comes from the learned admission decision itself, not merely from using a more generous static allocation: a static-at-cap policy set at the same maximum still collapses under sustained load in over a quarter of its episodes, while DQN never collapsed across a reverification of 25 episodes. We also report the control-loop latency and model footprint needed for practical deployment, and describe a calibration issue discovered and corrected in our QoE estimation during testing. A follow-up offline experiment under heavier, congested multi-slice traffic shows DQN learns to prioritise URLLC at the expense of the other two slices, consistent with the URLLC-first objective of our earlier simulated work, though this comes at a net cost once each slice's declared importance is taken into account. These findings demonstrate the feasibility, and surface the remaining gaps, of deploying DRL-driven SAC on real Open RAN hardware.

Index Terms—5G, Network Slicing, Slice Admission Control, Deep Reinforcement Learning, Quality of Experience, Open RAN

I. INTRODUCTION AND BACKGROUND

Network slicing (NS) lets a single physical 5G deployment serve heterogeneous service classes under per-slice Service Level Agreements (SLAs): Ultra-Reliable Low-Latency Communication (URLLC) requires sub-millisecond latency and high reliability, Enhanced Mobile Broadband (eMBB) demands high throughput, and Massive Machine-Type Communication (mMTC) accommodates large numbers of low-power devices. Slice Admission Control (SAC) decides whether an incoming slice request can be accommodated given current resource availability. Static, threshold-based SAC cannot adapt its allocation as demand shifts, motivating context-aware, learning-based approaches.

Our prior work applied Deep Reinforcement Learning (DRL) to SAC under gNodeB (gNB) congestion, using SLA-

driven reward shaping to prioritise URLLC traffic, and later extended this to a third slice (mMTC) with a load-balancing objective across multiple gNBs [1], [2]. Both studies, along with the majority of DRL-driven SAC work surveyed in our systematic review of the 2024–2026 literature [3], were validated entirely in simulation. This paper closes that gap for our own formulation: we realise the same admission-control problem on a real OAI gNB over its native E2 control interface, with the radio channel still simulated (rfsim), and centre the evaluation on Quality of Experience (QoE) rather than SLA compliance alone.

This paper makes the following contributions:

- 1) We deploy a DRL-based SAC policy on a real Open RAN E2 control loop and identify a structural difference from our earlier simulated work: the real E2 interface exposes no per-flow accept/reject primitive, only a per-slice resource ceiling, so admission must be realised as an adjustment of that ceiling.
- 2) We compare a QoE-aware reward directly against the SLA-only reward of [1], [2] on the same live hardware.
- 3) We add a static-at-cap arm that isolates whether the learned policy's advantage is genuine admission timing, or simply the result of using a more generous static allocation.
- 4) We report the engineering numbers (control-loop latency, model size) a real deployment needs, and a calibration issue found and fixed in our QoE estimation along the way.
- 5) We evaluate a congested, multi-slice traffic scenario offline and show the learned policy's tendency to prioritise URLLC carries over from simulation to this harder setting.

The remainder of this paper is organised as follows: Section II reviews related work on DRL-driven SAC. Section III describes the live testbed, the problem formulation, and the reward function. Section IV presents and discusses the results. Section V concludes the paper and outlines future work.

II. LITERATURE REVIEW

TODO(MANOJ): Literature review to be added here, following the style of Section II in our prior conference papers [1], [2] (narrative summary of ~10–12 related DRL-driven SAC / real-testbed studies, each introduced with “The authors in [X] proposed...”), concluding with a short paragraph positioning this paper against that related work. Candidate sources are already screened in our systematic review [3].

III. METHODOLOGY

A. Testbed

We deploy a real OAI gNB (ORANSlice fork [4]) with a simulated radio channel (rfsim), an Open5GS core, and three `nr-uesoftmodem` user equipment (UE), one per slice. The gNB has a nominal capacity of 100 Physical Resource Blocks (PRBs). Unlike our earlier simulated work, the real O-RAN E2 interface does not expose a binary accept/reject action; its only control primitive is a per-slice PRB ratio ceiling, enforced by the MAC scheduler [5]. We therefore realise the admission decision as a ceiling adjustment: an accept decision raises a slice’s ceiling toward a calibrated maximum, and a reject decision lowers it toward a floor. Ceiling values were calibrated by direct measurement on the live rig rather than assumed, since the ratio scale does not map linearly onto real PRBs at this cell’s configuration. The static baseline holds its ceiling fixed for the whole episode; a second static arm, *static-at-cap*, instead pins every slice’s ceiling at its calibrated maximum from the start, isolating whether a more generous fixed allocation alone can match a learned policy. Learned arms are DQN policies [6], trained offline against a closed-loop simulated traffic source, then evaluated live with frozen weights.

B. Problem Formulation

The gNB state at time t is defined as

$$s_t = \left(\frac{U_s(t)}{B}, C_t, \frac{L_s(t)}{L_{\max}} \forall s \in \mathcal{S} \right) \quad (1)$$

where $U_s(t)$ is the resource allocation of slice $s \in \mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$, B is the total available PRBs, C_t is the current congestion level, and $L_s(t)$ is slice s ’s queue length, normalised by its maximum capacity $L_{\max}$. The agent’s action at each step is the ceiling adjustment described above. The reward function is

$$r(s_t, a_t) = \sum_{s \in \mathcal{S}} (\omega_s R_s - \lambda_s) - \mu C_t \|a_t\|_1 + \eta \text{MOS}_t \quad (2)$$

where ω_s and λ_s are slice s ’s reward weight and SLA-violation penalty, R_s is the reward for serving slice s , and $\mu C_t \|a_t\|_1$ penalises total accepted volume by the current congestion level, matching the SLA-driven reward structure of our earlier work [1], [2]. Setting the QoE weight $\eta = 0$ recovers the SLA-only reward used in that prior work; setting $\eta = 1$ adds a Mean Opinion Score (MOS) term, estimated by a calibrated QoE-scoring model, giving the QoE-aware reward evaluated here.

TABLE I
LIVE SLA COMPLIANCE (%), MEAN OVER 3 SEEDS

Arm	eMBB	URLLC	mMTC
Static baseline	73.7	73.4	73.8
Static-at-cap	83.1	73.4	73.9
DQN (SLA reward)	100.0	100.0	100.0
DQN (QoE reward)	100.0	100.0	100.0

Both variants are compared on the same live hardware. We also report a priority-weighted compliance score, using each slice’s declared importance weight rather than an unweighted average across slices of unequal importance.

C. Evaluation Setup

Each arm was evaluated live across 3 random seeds and 5 episodes per seed (45 episodes for the two learned arms, 15 each for the static baseline and the static-at-cap arm), with frozen policy weights. The static-at-cap arm reuses the static baseline’s own control code unmodified, with its ceiling parameter changed only in configuration. Training was performed offline, over 300 episodes per seed, against a closed-loop simulated traffic source before live deployment.

IV. RESULTS AND DISCUSSION

A. Live Testbed Results

Table I summarises SLA compliance for all four arms across all 45 (learned) or 15 (static) live episodes. DQN sustains 100% SLA compliance on every slice, in every episode, under both the SLA-only and QoE-aware reward, against the static baseline’s seed-dependent 73.6% (worst single episode 0%). Fig. 1 shows why: the baseline’s commanded ceiling stays flat at its starting value for the whole episode, while DQN raises each slice’s ceiling toward its calibrated maximum within the first few steps and holds it there.

A natural question is whether DQN’s advantage simply reduces to using a more generous fixed ceiling. The static-at-cap arm rules this out: despite starting every episode already at the calibrated maximum, it collapses into the same severe backlog condition the static baseline shows in 4 of its 15 episodes (27%), while DQN, re-evaluated across a further round of 25 episodes on 5 different seeds, never collapsed once (a difference confirmed significant by a Fisher exact test, $p = 0.0149$). Both non-learning arms fail for the same reason: neither ever rejects a request under backlog pressure. DQN’s contribution is learning *when* to reject, which no fixed ceiling can do.

Fig. 2 shows this episode-by-episode: DQN sits at exactly 100% in every episode under both reward variants, while the static baseline and the static-at-cap arm each show the same bimodal split between fully-compliant and collapsed episodes.

B. Deployment Feasibility

Measured directly against the real gNB, the control loop’s E2 message round trip has a median latency of 0.57 ms, and the DQN model itself (under 20,000 parameters) selects an action in under 70 μs on CPU. Both are a small fraction of

Commanded PRB ceiling: baseline vs. best learned arm

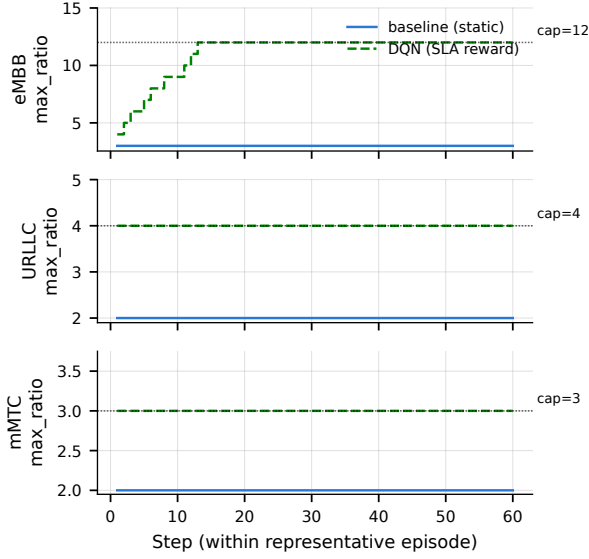


Fig. 1. Commanded ceiling per slice over one representative episode: static baseline (flat) vs. DQN (rises to its calibrated maximum).

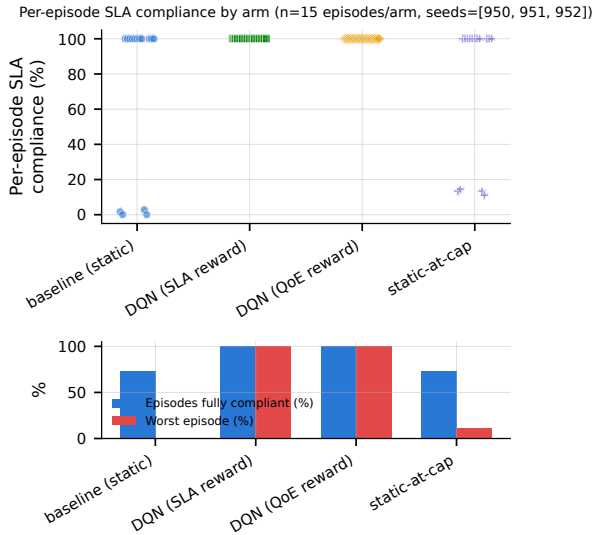


Fig. 2. Per-episode SLA compliance (%). Top: one point per episode. Bottom: fraction of episodes fully SLA-compliant and worst single-episode compliance. Both non-learning arms show the same bimodal split.

the 5-second control interval used in this study, indicating the learned policy adds negligible overhead to the control loop.

Comparing offline training predictions against the live results showed a good match for eMBB traffic, but not for URLLC and mMTC, which were trained under a different demand assumption than what was observed live. During this comparison we also identified a calibration issue in our QoE estimation: the throughput measure used to estimate MOS during training did not match the units produced by the live environment, which understated MOS for well-served traffic. We corrected this calibration and verified it against real observed operating points; a fully powered live re-evaluation

TABLE II
SLA COMPLIANCE (%) UNDER CONGESTED, RANDOMISED TRAFFIC (OFFLINE)

Arm	URLLC	eMBB	mMTC
Static baseline	22.6	34.5	19.0
DQN (SLA reward)	30.9	7.9	9.2
DQN (QoE reward)	27.0	8.2	10.8

under the corrected calibration is ongoing.

C. Congested Scenario (Offline)

Sections IV-A and IV-B use one constant demand level per episode. To test a harder, more realistic scenario, we evaluate the same trained policies offline under continuously varying, randomised multi-slice traffic with genuine cross-slice resource contention: when combined demand exceeds the shared PRB pool, every slice's served allocation is scaled down in proportion to its request, rather than each slice being served independently. Table II reports SLA compliance under this scenario for a held-out set of episodes.

Under both reward variants, DQN raises URLLC compliance above the static baseline (22.6% $\rightarrow$ 30.9% for the SLA-only reward, $\rightarrow$ 27.0% for the QoE-aware reward) by actively trading off eMBB and mMTC compliance – the same URLLC-first behaviour our earlier simulated work targeted [1], [2], now shown under real cross-slice contention rather than each slice's demand being satisfied independently. This tradeoff, however, is not a net win once each slice's declared importance is taken into account: weighted by each slice's priority, the static baseline still scores higher overall (27.2% vs. 21.0% for DQN with the SLA reward and 19.0% with the QoE reward), because eMBB's declared importance is high enough that the compliance it gives up outweighs URLLC's gain. The tradeoff is real and repeatable, but whether it or the underlying priority weights should be revisited is an open question for future work. This result is offline only and has not yet been reproduced on the live rig.

D. Summary of Findings

The live results confirm that a learned admission policy clearly outperforms static control under constant load, and that this advantage is specifically due to the learned admission decision, not a more generous static allocation. The offline congested scenario shows the same policy can learn to prioritise URLLC under harder, more realistic contention, consistent with our earlier simulated findings, though this is not yet an unambiguous improvement once every slice's relative importance is accounted for. Together, these results narrow the gap between our earlier simulation-only studies and a real deployment, while surfacing concrete next steps: a calibration fix still to be fully re-validated live, and a congested scenario still awaiting live reproduction.

V. CONCLUSION AND FUTURE WORK

This paper realised the DRL-based slice admission control problem from our prior simulated work [1], [2] on a real OAI gNB over its native E2 control loop, with the radio channel

still simulated. Across 45 live episodes, DQN sustained 100% SLA compliance under both the SLA-only and QoE-aware reward, against a static baseline's seed-dependent 73.6%. A static-at-cap arm confirmed this advantage comes from the learned admission decision itself: it collapsed at the same rate as the static baseline despite starting every episode at the best possible fixed allocation, while DQN never collapsed across a reverified 25 episodes. The control loop adds negligible latency and model overhead, supporting the feasibility of real-time deployment. A calibration issue in our QoE estimation was identified and corrected, and an offline evaluation under congested, randomised multi-slice traffic showed the learned policy can prioritise URLLC consistent with our earlier simulated work, though not yet as a clear win once slice priorities are weighted.

Our future work will (1) complete a fully powered live re-evaluation under the corrected QoE calibration, (2) reproduce the congested, multi-slice scenario on the live rig rather than offline, and (3) extend this single-agent, single-gNB formulation toward the multi-agent, multi-gNB setting proposed in our systematic review [3].

ACKNOWLEDGMENT

This research was supported by the Ministry of Higher Education (MOHE), Malaysia through the Fundamental Research Grant Scheme (FRGS/1/2024/ICT06/TAYLOR/02/1).

REFERENCES

- [1] M. P. Kunasegran, W. L. Pang, and S. K. Phang, "Optimising resource allocation in 5G networks: Balancing URLLC and eMBB traffic under gNB congestion," in *2025 Multimedia University Engineering Conference (MECON)*, 2025, pp. 1–6.
- [2] —, "SLA-aware DRL for joint slice admission control and load balancing in multi-service 5G RAN," in *2025 9th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, 2025, pp. 207–212.
- [3] —, "Comprehensive review of optimisation techniques for user-centric distributed network slicing in 5G networks," *IEEE Access*, 2026, vERIFY: final volume/issue/DOI once formally published – title page seen locally still shows placeholder publication-date/DOI fields.
- [4] WinesLab, "ORANSlice: O-RAN network slicing on OpenAirInterface," <https://github.com/wineslab/ORANSlice>, oRANSlice fork, OAI 2024.w28 base, commit b9bcc9b. VERIFY: confirm citation format WinesLab/Northeastern prefers for this repo.
- [5] O-RAN Alliance, "O-RAN working group 3, near-RT RIC architecture and E2 general aspects and principles (E2AP)," O-RAN Alliance, Tech. Rep., 2023, vERIFY: exact spec number/version (e.g. O-RAN.WG3.E2AP-v03.00) and year.
- [6] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.